

NASA SPACE APPS CHALLENGE 2026 · BANGLADESH REGION · PRESCREENING ENTRY

# FLARE-X: 240 Seconds of Glory

Official Prescreening Video Narration Script, Screen Directing Plan & Eligibility Verification

CHALLENGE CATEGORY

Flame in Freefall

TEAM IDENTITY

Team Alpine Ibex

SPOKEN DURATION

3:35 – 3:40 (220.0s)

SAFETY BUFFER

20.0s Below Limit

## 👥 Official Team Identification & Space Apps Roster

All team members are registered participants from Bangladesh. Meets full eligibility rules with **+5% Women Participation Bonus**.

| FULL NAME           | SPACE APPS HANDLE | OFFICIAL ROLE & MISSION FOCUS                    | EVENT / LOCATION |
|---------------------|-------------------|--------------------------------------------------|------------------|
| Soyebuzaman Naim    | @soyebuzamannaim  | Full Stack Developer                             | Bangladesh       |
| Saber Hossain Mahim | @sabermahim       | AI ML Engineer                                   | Bangladesh       |
| Abdullah Al Masum   | @abdullahalmasum  | Researcher                                       | Bangladesh       |
| Mahzabin Muntaha    | @muntaha02        | UI UX Designer (Women Participation Bonus (+5%)) | Bangladesh       |
| Hamza               | @hamza            | Backend Developer & Researcher                   | Bangladesh       |
| Nahid               | @nahid            | Video Editor                                     | Bangladesh       |

## 🎬 Second-by-Second Video Directing Plan: When to Show What

Structured strictly around the 4-Quadrant NASA Space Apps Bangladesh "240 Seconds of Glory" Pitch Model.

### Quadrant 01 — WHO: Attention & Authenticity (0:00 – 0:38 | 38s)

Rubric Checklist: Spend first 45s grabbing attention · Win with human story · First 15s circular hook · Name team members out loud.

| TIMECODE    | WHAT TO SHOW ON SCREEN (VISUAL SOURCE)                                                         | ACTION & CAMERA CHOREOGRAPHY                                                                                                                                                                                        |
|-------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0:00 – 0:08 | Slide 1:<br>slide_01_opener_microgravity.svg<br>(Destiny Module Cutaway & Astronaut)           | Slow cinematic push into ISS Destiny module pressure hull cutaway. Soft red pulse alert.<br><b>On-Screen:</b> ON EARTH, YOU RUN OUTSIDE. IN SPACE, YOU ARE LOCKED IN WITH YOUR FIRE.                                |
| 0:08 – 0:18 | Slide 1 Centerpiece: Blue Spherical Flame & Duct                                               | Reticle crosshair zooms on duct airflow callout; blue spherical diffusion flame creeping backward into ventilation ducts.<br><b>On-Screen:</b> Buoyancy Vanishes. Silent Creeping Spheres Inside Ventilation Ducts. |
| 0:18 – 0:38 | Slide 2: slide_02_team_roster.svg<br>(Featuring Official Poster<br>team_intro_alpine_ibex.png) | Cut cleanly to Slide 2. Left side displays the official Alpine Ibex poster; right side shows all 6 team badges: Naim, Mahim, Masum, Muntaha (Women Participation Bonus (+5%)), Hamza, and Nahid.                    |

### 🗣️ SPOKEN VOICEOVER (0:00 – 0:38 | 77 WORDS · TONE: URGENT, VISCERAL, ENGAGING)

"Imagine a single electrical spark on a spacecraft.

Without gravity, the flame doesn't rise. It forms a silent, hovering blue sphere. Feeding on the cabin's oxygen, it creeps slowly across surfaces and straight into the air ducts.

Three hundred miles above Earth, you can't run outside. You are locked in with it.

To protect Artemis crews, NASA conducted more than 1,500 fire experiments in orbit. We are Team Alpine Ibex from Bangladesh: Naim, Mahim, Masum, Muntaha, Hamza, and Nahid. And this is FLARE-X."

## Quadrant 02 — WHY: Create Empathy for the Problem (0:38 – 1:30 | 52s)

Rubric Checklist: Humanize problem for Artemis crew · Killer data point (Over 1,500 flight burns) · AI hallucinations · Under 60s rule.

| TIMECODE    | WHAT TO SHOW ON SCREEN (VISUAL SOURCE)                                                                         | ACTION & CAMERA CHOREOGRAPHY                                                                                                                                                                                                                                |
|-------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0:38 – 0:52 | <b>Slide 3 (Left):</b><br>slide_03_research_bottleneck.svg<br>(Engineer searching unstructured reports & PDFs) | Show aerospace engineer typing query: "Will PMMA ignite at 18% O <sub>2</sub> ?" Pan across unstructured 300-page NASA Technical Report PDFs and raw PSI spreadsheets.<br><b>On-Screen:</b> OVER 1,500 FLIGHT BURNS BURIED IN UNSTRUCTURED NASA DATA & PDFs |
| 0:52 – 1:10 | <b>Slide 3 (Right):</b> Generic AI Strike-Through                                                              | Crimson red slash cuts across generic LLM hallucinating fake flame speeds. Space safety cannot tolerate guesswork.<br><b>On-Screen:</b> GENERIC AI HALLUCINATES FLAME PHYSICS · SAFETY DEMANDS PROOF                                                        |
| 1:10 – 1:30 | <b>Slide 3 (Bottom):</b> 5 Isolated Combustion Silos                                                           | 5 discrete columns appear: Solids (BASS-II), Droplets (FLEX), Gas (SPICE), Spacecraft (SAFFIRE), Smoke (SAME).                                                                                                                                              |

### SPOKEN VOICEOVER (0:38 – 1:30 | 98 WORDS · TONE: CLEAR, SHARP, THOUGHTFUL)

"When Artemis engineers choose materials for a new Moon habitat at eighteen percent oxygen, where do they look? NASA's fire data is buried inside three-hundred-page PDFs, unstructured spreadsheets, and raw flight archives.

Worse, if you ask a standard AI chatbot, it just invents numbers. It doesn't understand fire physics. A burning liquid droplet in FLEX doesn't behave like a solid wall panel in BASS-Two. Different materials, different airflows, different pressures.

In spaceflight, guessing is fatal. Researchers don't need a bot that guesses. They need a tool that finds real NASA flight tests, checks safety boundaries, and proves where every answer comes from."

### Quadrant 03 — WHAT: Your Big Idea — Explain Your Innovation (1:30 – 2:20 | 50s)

*Rubric Checklist: Detail core concept · 6-step workflow · Real machine learning with flight validation · Safety Envelope Guard.*

| TIMECODE    | WHAT TO SHOW ON SCREEN (VISUAL SOURCE)                                        | ACTION & CAMERA CHOREOGRAPHY                                                                                                                                                                           |
|-------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1:30 – 1:50 | <b>Slide 4:</b> slide_04_pipeline_routing.svg (6-Step Closed-Loop Pipeline)   | 6-step closed-loop pipeline lights up left to right on clean white cards: 01 Define → 02 Identify → 03 Retrieve → 04 Model → 05 Guard → 06 Trace.<br><b>On-Screen:</b> CLOSED-LOOP SCIENTIFIC WORKFLOW |
| 1:50 – 2:05 | <b>Slide 5:</b> slide_05_model_specialization.svg (Solid Polymer Family Card) | Spotlight on Card 3 (BASS-GB Solid Polymers). Highlight the verified benchmark:<br><b>On-Screen:</b> 79.31% Grouped Cross-Validation Accuracy (+27.6% Baseline Lift)                                   |
| 2:05 – 2:20 | <b>Slide 6:</b> slide_06_envelope_guard.svg (Hexagonal Shield Convex Hull)    | Display pulsing Section 7 Envelope Guard hexagonal shield with 6 convex hull dimensions.<br><b>On-Screen:</b> FLARE-X KNOWS WHEN NOT TO PREDICT · ZERO-EXTRAPOLATION PROTOCOL                          |

#### SPOKEN VOICEOVER (1:30 – 2:20 | 104 WORDS · TONE: CONFIDENT, CRISP ENGINEERING PRECISION)

"That is why we built FLARE-X. It takes any space cabin scenario through a simple, six-step scientific workflow.

First, it matches the material to real NASA flight campaigns—like BASS, FLEX, and SAFFIRE. Then, it searches our database of real flight tests and pulls up the closest experiments NASA ever conducted.

For solid cabin materials, our machine learning model predicts whether a flame will keep spreading, flicker unstably, or put itself out—with an honest seventy-nine percent accuracy backed by flight data.

And here is our most important rule: the Safety Envelope Guard. Before returning any prediction, it checks: Did NASA actually test these conditions in space?"

### Quadrant 04 — HOW: Live Demo, Prototype & Future Impact (2:20 – 3:40 | 80s)

*Rubric Checklist: Show working software · Reveal the 0.5s Envelope Refusal · State NASA open data out loud · Tantalize future vision.*

| TIMECODE    | WHAT TO SHOW ON SCREEN (VISUAL SOURCE)                                     | ACTION & CAMERA CHOREOGRAPHY                                                                                                                                                                                                                                                                                                              |
|-------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2:20 – 2:35 | <b>Live HTML Workbench (Tab 1):</b> project-page/index.html#demo-workbench | <b>SWITCH DIRECTLY TO LIVE BROWSER WINDOW.</b> Show 2D Experimental Support Map canvas with 145 real NASA spaceflight burns and white exploration star (★) at (21.0% O <sub>2</sub> , 15 cm/s).                                                                                                                                           |
| 2:35 – 2:52 | <b>Live HTML Workbench (Tab 2):</b> Real-Time Oxygen Sweep Slider          | Click [ Tab 2: Oxygen Sweep ]. Drag slider from 21% down to 18%. UI reacts instantly: status becomes QUENCHED, burn duration 42.0s, nearest test updates live to BASS-II Test 31 (Note: On-screen UI explicitly displays authentic acronym BASS-II while spoken teleprompter reads BASS-Two for smooth narration).                        |
| 2:52 – 3:00 | <b>Live HTML Workbench (Tab 2):</b> Slider Sweeps to 13.0% O <sub>2</sub>  | Cursor smoothly drags slider into unvalidated territory: 13.0% O <sub>2</sub> .                                                                                                                                                                                                                                                           |
| 3:00 – 3:15 | <b>THE FLAGSHIP ABSTENTION MOMENT:</b> Red Hazard Refusal Barrier          | <b>PAUSE 0.5s IN AUDIO.</b> Prediction card vanishes into crimson hazard border:<br> <b>OUTSIDE NASA MODEL ENVELOPE — PREDICTION WITHHELD</b><br>Reason: Local support radius violated (min O <sub>2</sub> ≥ 15.0%).<br>Withheld for astronaut safety. |

| TIMECODE    | WHAT TO SHOW ON SCREEN (VISUAL SOURCE)                                          | ACTION & CAMERA CHOREOGRAPHY                                                                                                                             |
|-------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3:15 – 3:30 | <b>Live HTML Workbench:</b><br>Provenance Drawer & DOI Links                    | Click Provenance Drawer on BASS-II Test 31. Drawer slides open revealing NASA Technical Report citation (NASA/TP-2016-219216) and PSI DOI link.          |
| 3:30 – 3:40 | <b>Slide 9:</b> slide_09_closing_end_card.svg<br>(Light Theme Closing End Card) | Smooth cut to Light Theme End Card. Displays FLARE-X logo, all 6 team credits, Alpine Ibex nominee badge, and NASA Space Apps 2026. Fade to clean white. |

 **SPOKEN VOICEOVER (2:20 – 3:40 | 147 WORDS · TONE: DEMONSTRATIVE, GROUNDED, POWERFUL)**

"Here is our live project in action. On this map, every single dot is a real fire NASA lit in space. The star is our test scenario.

Watch what happens when we lower oxygen from twenty-one percent down to eighteen percent. The system updates live. It finds the nearest flight test from BASS-Two, and shows that the flame puts itself out in forty-two seconds.

Now, let's push oxygen down to thirteen percent.

**[0.5-SECOND DRAMATIC PAUSE / LET JUDGES ABSORB THE RED REFUSAL SCREEN]**

Look at the screen. FLARE-X stopped. It did not guess. It triggered an Envelope Refusal, telling us: NASA never tested this material at thirteen percent oxygen, so for astronaut safety, we will not predict.

Every single result links directly to primary NASA research papers and official dataset DOIs. Because on the way to the Moon and Mars, untested data is unsafe data.

We are Team Alpine Ibex. Safe fire science for the next frontier."

## Master Continuous Teleprompter Script (426 Words)

Full spoken script in plain, natural human English. Total runtime is strictly **3:30 to 3:35 (212.0s spoken + dramatic pauses)** at 121–128 words per minute.

### [0:00 - QUADRANT 01: WHO · ATTENTION & AUTHENTICITY]

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### [0:38 - QUADRANT 02: WHY · CREATE EMPATHY FOR THE PROBLEM]

When Artemis engineers choose materials for a new Moon habitat at eighteen percent oxygen, where do they look? NASA's fire data is buried inside three-hundred-page PDFs, unstructured spreadsheets, and raw flight archives.

Worse, if you ask a standard AI chatbot, it just invents numbers. It doesn't understand fire physics. A burning liquid droplet in FLEX doesn't behave like a solid wall panel in BASS-Two. Different materials, different airflows, different pressures.

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### [1:30 - QUADRANT 03: WHAT · YOUR BIG IDEA: INNOVATION]

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First, it matches the material to real NASA flight campaigns—like BASS, FLEX, and SAFFIRE. Then, it searches our database of real flight tests and pulls up the closest experiments NASA ever conducted.

For solid cabin materials, our machine learning model predicts whether a flame will keep spreading, flicker unstably, or put itself out—with an honest seventy-nine percent accuracy backed by flight data.

And here is our most important rule: the Safety Envelope Guard. Before returning any prediction, it checks: Did NASA actually test these conditions in space?

### [2:20 - QUADRANT 04: HOW · DEMO, PROTOTYPE & IMPACT]

Here is our live project in action. On this map, every single dot is a real fire NASA lit in space. The star is our test scenario.

Watch what happens when we lower oxygen from twenty-one percent down to eighteen percent. The system updates live. It finds the nearest flight test from BASS-Two, and shows that the flame puts itself out in forty-two seconds.

Now, let's push oxygen down to thirteen percent.

**[0.5-SECOND SILENCE]**

Look at the screen. FLARE-X stopped. It did not guess. It triggered an Envelope Refusal, telling us: NASA never tested this material at thirteen percent oxygen, so for astronaut safety, we will not predict.

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## ✓ NASA Space Apps Bangladesh 2026 Eligibility Verification

- ✓ **Under 240-Second Rule:** Verified runtime 220.0s (3:40 total) leaves a safe 20.0s buffer below the disqualification cutoff.
- ✓ **Mandatory Team Roll Call:** Names every single member out loud: Soyebuzaman Naim, Saber Hossain Mahim, Abdullah Al Masum, Mahzabin Muntaha, Hamza, and Nahid.
- ✓ **Women Participation Bonus (+5%):** Mahzabin Muntaha leads UI/UX design and technical experience architecture.
- ✓ **NASA Open Data Out Loud:** Cites NASA Physical Sciences Informatics (PSI), BASS, BASS-II, FLEX, SAFFIRE, and NTRS papers.
- ✓ **Working Prototype Demonstrated:** Real interactive browser screen-recording in Block 4 directly showing live data recalculation.
- ✓ **English Closed Captions:** Synchronized broadcast-standard SubRip (.srt) track provided in repository.
- ✓ **Under-18 Safety Rule:** Verified 100% compliant with no minor likenesses, voices, or names.
- ✓ **Open Source Licensing:** Apache-2.0 open-access repository hosted publicly on GitHub.